

5(a)(i)	$Q_A = CV$	A1
5(a)(ii)	$E_A = \frac{1}{2}CV^2$	A1
5(b)(i)	some of the charge transfers to (the plates of) capacitor B	B1
	transfer is because the p.d.s across the capacitors are not equal or transfer stops when the p.d.s across the capacitors become equal	B1
5(b)(ii)	$V_A = V_B$	M1
	charge on A + charge on B = CV	M1
	$CV_B + 3CV_B = CV$ leading to $V_B = V/4$	A1
	or	
	$C_T = 4C$	(M1)
	$Q_T = CV$	(M1)
	$V_B = CV/4C = V/4$	(A1)
5(b)(iii)	$\Delta E = \frac{1}{2}CV^2 - nCV^2$, where n is a multiple that is less than $\frac{1}{2}$ or total final energy = $\frac{1}{2} \times 4C \times (V/4)^2$ $= \frac{1}{8}CV^2$	C1
	$\Delta E = \frac{1}{2}CV^2 - \frac{1}{8}CV^2$ $= \frac{3}{8}CV^2$	A1